

Dimension estimates for C^1 iterated function systems

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Preliminaries: IFS and attractors

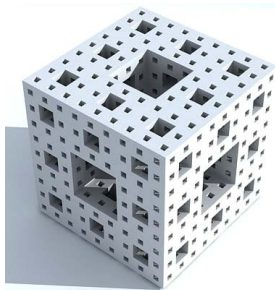
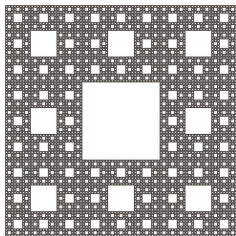
- Consider a family of contracting maps $\{f_i\}_{i=1}^{\ell}$ on $\mathbb{R}^d$, which is called an **iterated function system** (IFS).
- By Hutchinson (1981), there is a unique nonempty compact set $K \subset \mathbb{R}^d$ such that

$$K = \bigcup_{i=1}^{\ell} f_i(K).$$

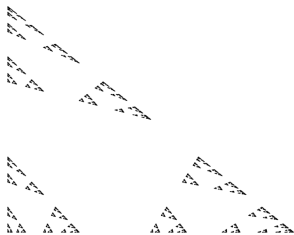
The set K is called is the **attractor** of the IFS $\{f_i\}_{i=1}^{\ell}$.

- K is said to be **self-similar** if all f_i are similarity maps (i.e. $f_i(x) = \rho_i U_i x + a_i$, with $0 < \rho_i < 1$ and U_i being orthogonal); **self-affine** if all f_i are affine maps; and **self-conformal** if all f_i are conformal maps.

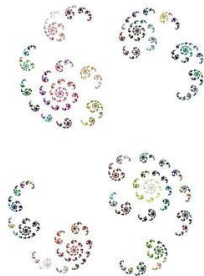
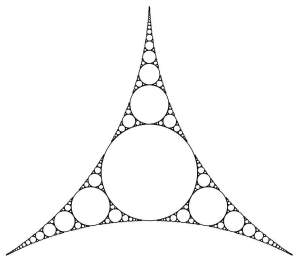
Examples—self-similar sets



Examples—self-affine sets



Examples—self-conformal sets



A fundamental question in fractal geometry and dynamical systems

Question: How to estimate and compute the variant fractal dimensions (e.g. Hausdorff and box-counting dimensions) of the attractor K of an IFS?

- Given $E \subset \mathbb{R}^d$ and $s \geq 0$, define

$$\mathcal{H}_\infty^s(E) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(A_i)^s : E \subset \bigcup_{i=1}^{\infty} A_i \right\}.$$

We call $\mathcal{H}_\infty^s(E)$ the s -dimensional Hausdorff content of E .
The Hausdorff dimension of E is

$$\dim_H E = \inf \{s \geq 0 : \mathcal{H}_\infty^s(E) = 0\}.$$

- The box-counting dimension of E is defined by

$$\dim_B E = \lim_{r \rightarrow 0+} \frac{\log N_r(E)}{\log(1/r)},$$

provided the limit exists, where $N_r(E)$ is the smallest number of balls with radius r which can cover E .

Dimension of self-similar and self-conformal sets

- (Non-overlapping self-similar case) Moran 1946 & Hutchinson 1981: If f_i 's are **similarities** + **open set condition** (OSC), then

$$\dim_H K = \dim_B K = s, \quad \text{with } \sum_{i=1}^{\ell} r_i^s = 1,$$

where r_i is the contraction ratio of f_i .

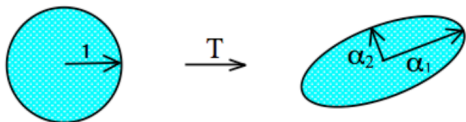
- (Overlapping self-similar case) Significant results by Hochman 2014, Varjú 2019, Rapaport 2020, Rapaport-Varjú 2021.
- Similar result holds for C^1 **conformal IFSs with OSC** (Bowen 1979, Ruelle 1981): the dimension s is given by the Bowen equation:

$$P(s \log \|D_{\pi \sigma x} f_{x_1}\|) = 0,$$

where P stands for the **topological pressure**.

Falconer's work on self-affine sets (1988)

- For a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, let $\alpha_1 \geq \alpha_2$ be the **singular values** of T , i.e. the semi-axis lengths of $T(\text{unit ball})$.



The **singularity function** of T is defined by

$$\phi^s(T) = \begin{cases} \alpha_1^s & \text{if } 0 \leq s \leq 1 \\ \alpha_1 \alpha_2^{s-1} & \text{if } 1 < s \leq 2 \\ (\alpha_1 \alpha_2)^{s/2} & \text{if } s > 2. \end{cases}$$

- More generally, for $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, define

$$\phi^s(T) = \begin{cases} \alpha_1 \dots \alpha_{j-1} \alpha_j^{s-j+1} & \text{for } j-1 < s \leq j \leq d, \\ (\alpha_1 \dots \alpha_d)^{s/d} & \text{for } s > d. \end{cases}$$

- $\mathcal{H}_\infty^s(T(\text{unit ball})) \approx_{s,d} \phi^s(T)$ for non-integer $s \in (0, d)$, where $\mathcal{H}_\infty^s$ stands for s -dimensional Hausdorff content.
- (Sub-multiplicativity): $\phi^s(AB) \leq \phi^s(A)\phi^s(B)$.

Falconer's work on self-affine sets (1988)

Define

$$s(T_1, \dots, T_m) := \inf \left\{ s \geq 0 : \sum_{n=1}^{\infty} \sum_{i_1 \dots i_n} \phi^s(T_{i_1 \dots i_n}) < \infty \right\},$$

which is called the **affinity dimension** of K .

Theorem (Falconer 1988, Solomyak 1998)

Let K be the attractor of an affine IFS $\{T_i x + a_i\}_{i=1}^m$ on $\mathbb{R}^d$. Then

$$\dim_H(K) \leq \overline{\dim}_B(K) \leq s(T_1, \dots, T_m),$$

Assume $\|T_i\| < 1/2$ for all i . Then for $\mathcal{L}^{md}$ -a.e. $(a_1, \dots, a_m)$,

$$\dim_H(K) = \dim_B(K) = \min\{d, s(T_1, \dots, T_m)\}.$$

Ideas of Falconer: upper bound

- The upper bound $\overline{\dim}_B K \leq s(T_1, \dots, T_m)$ follows from a clever box-counting argument.
- Alternatively, for the inequality $\dim_H K \leq s(T_1, \dots, T_m)$, one can choose a large $R > 0$ such that

$$T_i(B(0, R)) \subset B(0, R), \quad i = 1, \dots, m,$$

where $B(0, R)$ stands for the ball of radius R centred at the origin. Then use directly use the inequality

$$\mathcal{H}_\infty^s(T_{i_1 \dots i_n}(B(0, R))) \leq cR^s \phi^s(T_{i_1 \dots i_n}), \quad \forall i_1 \dots i_n$$

where $c > 0$ is independent of R and T_i . For $t > s(T_1, \dots, T_m)$, one has

$$\sum_n \sum_{i_1 \dots i_n} \mathcal{H}_\infty^t(T_{i_1 \dots i_n}(B(0, R))) < \infty,$$

which implies $\dim_H K \leq t$.

Ideas of Falconer: lower bound

- Let $t < s < s(T_1, \dots, T_m)$. First build a measure μ on $\Sigma = \{1, \dots, m\}^{\mathbb{N}}$ such that

$$\mu([i_1 \dots i_n]) \leq C' \phi^s(T_{i_1 \dots i_n}), \quad \forall i_1 \dots i_n$$

Then use the above inequality and Fubini Theorem to show that

$$\int_{B(0,R)} I_t(\pi_{\mathbf{a}}\mu) \, d\mathbf{a} < \infty, \quad \forall R > 0,$$

where $\pi_{\mathbf{a}}\mu$ is the push forward of μ by the coding map $\pi_{\mathbf{a}}$ associated to the IFS $\{T_i x + a_i\}$, and $I_t(\eta)$ is the t -energy of η defined by

$$I_t(\eta) := \int \int |x - y|^{-t} \, d\eta(x) d\eta(y).$$

Hence for a.e. $\mathbf{a}$, $I_t(\pi_{\mathbf{a}}\mu) < \infty$, which implies $\dim_H K \geq \dim \pi_{\mathbf{a}}\mu \geq t$.

Remark:

- Falconer's formula does hold for self-affine sets in $\mathbb{R}^2$ and $\mathbb{R}^3$ under some irreducibility and separation conditions.
([Barany-Hochman-Rapaport 2019](#), [Hochman-Rapaport 2022](#), [Rapaport 2022](#), [Morris-Sert 2023](#))
- There are some self-affine sets (e.g. Bedford-McMullen carpets) which do not satisfy Falconer's formula.

Our main goal

To understand the dimension of the attractors of general
non-linear, non-conformal C^1 IFSs.

Attractors of general C^1 IFSs: a dimensional upper bound

- Let $\{f_i\}_{i=1}^m$ be a C^1 IFS on $\mathbb{R}^k$ with attractor E . Given $s \geq 0$, define

$$P(s) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{i_1 \dots i_n} \sup_{x \in [i_1 \dots i_n]} \phi^s(D_{\pi \sigma^n x} f_{x|n})$$

The **singularity dimension** of E is defined as

$$\dim_S(E) = \inf\{s \geq 0 : P(s) = 0\}.$$

- Falconer** (1994): $\overline{\dim}_B(E) \leq \dim_S(E)$ under some strong assumptions (e.g. 1-bunched distortion + C^2) on the IFS.
- Yingjie Zhang** (1997): $\dim_H(E) \leq \dim_S(E)$.

Theorem (F.–Simon, 2020)

Let E be the attractor of a C^1 IFS $\{f_i\}_{i=1}^m$ on $\mathbb{R}^k$. Then

$$\overline{\dim}_B(E) \leq \dim_S(E).$$

Moreover, the packing dimension of any *ergodic stationary measure* on E is bounded above by its *Lyapunov dimension*.

- Difficulty: $f_{i_1 \dots i_n}(B(1))$ may not be comparable to $D_x f_{i_1 \dots i_n}(B(1))$.
- We extend the ideas of [Douady-Oesterlé \(1980\)](#) and [Y.J. Zhang \(1997\)](#), and use the (sub-additive) thermodynamic formalism.

Question: Do we have

$$\dim_H E = \dim_B E = \dim_S(E)$$

for the attractor E of some “typical” C^1 IFS?

To answer this question, we introduce a notion of [generalised transversality condition](#) (GTC), which was inspired by the work of Jordan-Pollicott-Simon (2007) who introduced the [self-affine transversality condition](#) for affine IFSs.

A generalised transversality condition (GTC)

Let $\ell \geq 2$ and let $\mathcal{F}^t = \{f_1^t, \dots, f_\ell^t\}$, $t \in \Omega$, be a parametrized family of C^1 IFSs defined on a common compact subset Z of $\mathbb{R}^d$, where (Ω, ρ) is a separable metric space, such that the following two conditions hold:

- (C1) The maps f_i^t have a common Lipschitz constant $\theta \in (0, 1)$, that is

$$|f_i^t(x) - f_i^t(y)| \leq \theta |x - y|$$

for all $1 \leq i \leq \ell$, $t \in \Omega$ and $x, y \in Z$.

- (C2) The mapping $t \mapsto f_i^t(x)$ is continuous over Ω for every given $x \in Z$ and $1 \leq i \leq \ell$.

For each $t \in \Omega$, let K^t denote the attractor of $\mathcal{F}^t$, and let $\Pi^t : \Sigma \rightarrow \mathbb{R}^d$ denote the coding map associated to the IFS $\mathcal{F}^t$. For $t \in \Omega$, $r > 0$ and $\mathbf{i} \in \Sigma_* := \bigcup_{n=0}^{\infty} \{1, \dots, \ell\}^n$, set

$$Z_{\mathbf{i}}^t(r) = \inf_{x \in \Sigma} \min \left\{ \frac{r^k}{\phi^k(D_{\Pi^t x} f_{\mathbf{i}}^t)} : k = 0, 1, \dots, d \right\},$$

where $f_{\mathbf{i}}^t := f_{i_1}^t \circ \dots \circ f_{i_n}^t$ for $\mathbf{i} = i_1 \dots i_n$.

Geometrically,

$$Z_{\mathbf{i}}^t(r) \approx \inf_{x \in \Sigma} \mathcal{L}^d \left(B(1) \cap (D_{\Pi^t x} f_{\mathbf{i}}^t)^{-1} B(r) \right),$$

where $B(r)$ stands for the ball of radius r centred at the origin.

Definition

Let η be a locally finite Borel measure on Ω . We say that the family $\mathcal{F}^t$, $t \in \Omega$, satisfies a **generalized transversality condition** (GTC) with respect to η if there exist $\delta_0 > 0$ and a function $\psi : (0, \delta_0) \rightarrow [0, \infty)$ with $\lim_{\delta \rightarrow 0} \psi(\delta) = 0$ such that the following statement holds: for every $t_0 \in \Omega$ and every $0 < \delta < \delta_0$, there exists a constant $C = C(t_0, \delta) > 0$ such that for all distinct $\mathbf{i}, \mathbf{j} \in \Sigma$ and $r > 0$,

$$\eta \{ t \in B(t_0, \delta) : |\Pi^t(\mathbf{i}) - \Pi^t(\mathbf{j})| < r \} \leq C e^{|\mathbf{i} \wedge \mathbf{j}| \psi(\delta)} Z_{\mathbf{i} \wedge \mathbf{j}}^{t_0}(r),$$

where $B(t_0, \delta)$ denotes the closed ball in Ω of radius δ centered at t , $\mathbf{i} \wedge \mathbf{j}$ denotes the common initial segment of $\mathbf{i}$ and $\mathbf{j}$, and $|\mathbf{i} \wedge \mathbf{j}|$ is the length of the word $\mathbf{i} \wedge \mathbf{j}$.

Theorem (F.–Simon, 2022)

Let $\mathcal{F}^t = \{f_1^t, \dots, f_\ell^t\}$, $t \in \Omega$, be a parametrized family of C^1 IFSs defined on a common compact subset Z of $\mathbb{R}^d$. Let η be a locally finite Borel measure on Ω . Assume that $(\mathcal{F}^t)_{t \in \Omega}$ satisfies the GTC with respect to η . Then for η -a.e. $t \in \Omega$,

$$\dim_H K^t = \dim_B K^t = \min\{d, \dim_S(K^t)\}.$$

Moreover, $\mathcal{L}^d(K^t) > 0$ for η -a.e. $t \in \{t' \in \Omega : \dim_S(K^{t'}) > d\}$.

Remark: The above theorem is a nonlinear analogue of the result of [Jordan-Pollicott-Simon](#) (2007) for affine IFSs. It also extends and generalizes the corresponding results of [Simon-Solomyak-Urbański](#) (2001) for $C^{1+\alpha}$ conformal IFSs on $\mathbb{R}$.

Now a natural question arises how to verify the GTC for a parametrized family of C^1 IFSs. In what follows we investigate this question for certain translational families of C^1 IFSs.

Definition

Let $\mathcal{F} = \{f_i\}_{i=1}^\ell$ be a C^1 IFS on a compact set $Z \subset \mathbb{R}^d$ such that $f_i(Z) \subset \text{int}(Z)$ for each i . Set

$$f_i^{\mathbf{t}} := f_i + \mathbf{t}_i, \quad i = 1, \dots, \ell,$$

where $\mathbf{t} = (\mathbf{t}_1, \dots, \mathbf{t}_\ell) \in \mathbb{R}^{\ell d}$ with $\mathbf{t}_i \in \mathbb{R}^d$. By continuity, there is a small $r_0 > 0$ such that $f_i^{\mathbf{t}}(Z) \subset \text{int}(Z)$ for every $\mathbf{t}$ with $|\mathbf{t}| < r_0$ and every i , where $|\cdot|$ is the Euclidean norm. Set $\mathcal{F}^{\mathbf{t}} = \{f_i^{\mathbf{t}}\}_{i=1}^\ell$ for each $\mathbf{t}$ with $|\mathbf{t}| < r_0$. We call $(\mathcal{F}^{\mathbf{t}})_{\mathbf{t} \in \Delta}$, where $\Delta := \{\mathbf{s} \in \mathbb{R}^{\ell d} : |\mathbf{s}| < r_0\}$, a **translational family of IFSs generated by $\mathcal{F}$** .

Definition

Let $\mathcal{F} = \{f_i\}_{i=1}^\ell$ be a C^1 IFS on a compact set $Z \subset \mathbb{R}^d$. We say that $\mathcal{F}$ is **dominated lower triangular**, if for each $z \in Z$ and $i \in \{1, \dots, \ell\}$, the Jacobian $D_z f_i$ of f_i at z is a lower triangular matrix such that

$$|(D_z f_i)_{11}| \geq |(D_z f_i)_{22}| \geq \dots \geq |(D_z f_i)_{dd}|.$$

Definition

Let $\ell \in \mathbb{N}$ with $\ell \geq 2$. Assume for $j = 1, \dots, n$, $\mathcal{F}_j = \{f_{i,j}\}_{i=1}^{\ell}$ is an IFS on $Z_j \subset \mathbb{R}^{q_j}$. Let $\mathcal{F} = \{f_i\}_{i=1}^{\ell}$ be an IFS on $Z_1 \times \dots \times Z_n \subset \mathbb{R}^{q_1} \times \dots \times \mathbb{R}^{q_n}$ given by

$$f_i(x_1, \dots, x_n) = (f_{i,1}(x_1), \dots, f_{i,n}(x_n)), \quad i = 1, \dots, \ell, \quad x_k \in Z_k.$$

We say that $\mathcal{F}$ is the **direct product** of $\mathcal{F}_1, \dots, \mathcal{F}_n$, and write $\mathcal{F} = \mathcal{F}_1 \times \dots \times \mathcal{F}_n$.

Theorem (F.–Simon, 2022)

Let $\mathcal{F} = \{f_i\}_{i=1}^\ell$ be a C^1 IFS on a compact convex set $Z \subset \mathbb{R}^d$ such that $f_i(Z) \subset \text{int}(Z)$ for each i . Suppose either one of the following 3 conditions holds:

(i) $\mathcal{F}$ is dominated lower triangular on Z satisfying

$$\max_{i \neq j} \left(\sup_{y, z \in Z} \|D_y f_i\| + \|D_z f_j\| \right) < 1. \quad (1)$$

(ii) $\mathcal{F}$ is a C^1 conformal IFS on Z satisfying (1).

(iii) $\mathcal{F} = \mathcal{F}_1 \times \cdots \times \mathcal{F}_n$, where for each $k \in \{1, \dots, n\}$, $\mathcal{F}_k$ is a C^1 IFS on a compact convex $Z_k \subset \mathbb{R}^{d_k}$ satisfying either (i) or (ii).

Then there is a small $r_0 > 0$ such that the translational family

$$\mathcal{F}^{\mathbf{t}} = \{f_i + \mathbf{t}_i\}_{i=1}^\ell, \quad \mathbf{t} = (\mathbf{t}_1, \dots, \mathbf{t}_\ell) \in \Delta := \{\mathbf{s} \in \mathbb{R}^{\ell d} : |\mathbf{s}| < r_0\},$$

satisfies the GTC with respect to the Lebesgue measure $\mathcal{L}^{\ell d}$ on Δ .

For the poof of this theorem, the most difficulty part is to deal with Case (i), i.e., when $\mathcal{F}$ is dominated lower triangular.

For a C^1 map $g : S \subset \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $z_1, \dots, z_d \in S$, we write

$$D_{z_1, \dots, z_d}^* g = \begin{bmatrix} \nabla^T g_1(z_1) \\ \vdots \\ \nabla^T g_d(z_d) \end{bmatrix} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1}(z_1) & \cdots & \frac{\partial g_1}{\partial x_d}(z_1) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d}{\partial x_1}(z_d) & \cdots & \frac{\partial g_d}{\partial x_d}(z_d) \end{bmatrix},$$

where g_i is the i -th component of the map g , $i = 1, \dots, d$.

A key technical part in the proof

The following proposition plays a key role in our proof.

Proposition

Let $\mathcal{F}^{\mathbf{t}}$, $\mathbf{t} \in \Delta$, be a translational family of IFSs generated by a dominated lower triangular C^1 IFS $\mathcal{F}$ on a compact subset S of $\mathbb{R}^d$. Then there exists a function $h : (0, r_0) \rightarrow (0, \infty)$ with $\lim_{\delta \rightarrow 0} h(\delta) = 0$ such that for each $\delta \in (0, r_0)$, there is $C(\delta) \geq 1$ so that

$$\|D_y f_{\omega}^{\mathbf{s}} \cdot (D_{z_1, \dots, z_d}^* f_{\omega}^{\mathbf{t}})^{-1}\| \leq C(\delta) e^{nh(\delta)} \quad (2)$$

for every $n \in \mathbb{N}$, $\omega \in \Sigma_n$, $y, z_1, \dots, z_d \in S$ and $\mathbf{s}, \mathbf{t} \in \Delta$ with $|\mathbf{s} - \mathbf{t}| < \delta$.

Besides, we also adopt and extend some ideas of [Falconer 1988](#), and [Jordan-Pollicott-Simon \(2007\)](#).

In the planar C^2 dominated lower triangular case, under the strong separation condition,

- [Falconer-Fraser-Lee 2021](#): L^q -spectra of stationary measures and box-counting dimension of the attractor.
- [Jurga-Lee 2022](#): Exact dimensionality of stationary measures.

Question 1. Is it true that for every C^1 IFS $\mathcal{F} = \{f_i\}_{i=1}^\ell$ on $\mathbb{R}^d$ satisfying (1), there is a neighborhood Δ of $\mathbf{0}$ in $\mathbb{R}^{\ell d}$ such that for $\mathcal{L}_{\ell d}$ -a.e. $\mathbf{t} = (\mathbf{t}_1, \dots, \mathbf{t}_\ell) \in \Delta$,

$$\dim_H K^{\mathbf{t}} = \dim_B K^{\mathbf{t}} = \min\{\dim_S K^{\mathbf{t}}, d\}?$$

where $K^{\mathbf{t}}$ is the attractor of the IFS $\mathcal{F}^{\mathbf{t}} = \{f_i + \mathbf{t}_i\}_{i=1}^\ell$.

Question 2. Do we have

$$\dim_H K = \dim_B K = \min\{\dim_S K, d\}$$

for the attractor K of a “generic” C^1 IFS $\mathcal{F}$ on $\mathbb{R}^d$ (in an appropriate sense)?

Thank you for listening !!!